Balancing performance, scale and circularity to guarantee recyclability in the future

Rajesh Balakrishnan, Advanced Materials, Aditya Birla Group, talks to JEC Composites Magazine about his group's portfolio and the importance of recyclability in all activities related to advanced materials. And even more so in the coming years.

Of all the companies involved in composite applications, Aditya Birla's Advanced Materials division is the one that is most closely engaged with the industry. The division sells resins and solutions for a variety of applications, including automotive, sporting goods, electronics and wind energy sectors. The second most important sector is the insulator and construction industry with composite rebars. Around 35-40% of the Advanced Materials division's international revenue comes from composites applications, mainly due to the resins business. Aditya Birla is also a leading player in the wind industry, holding around 80% of the Indian market.

Rajesh Balakrishnan explains: *"We have a strong position in wind energy, and it is a growing industry for us. In India, we currently have around 50 GW of wind energy capacity. According to the road map set out by the Indian government, this capacity is expected to reach 100 GW by 2030. We have various ongoing business relationships with major wind OEMs, including Suzlon, GE and INOX."* Rajesh also sees strong opportunities for the recyclability of wind blades because many countries are restricting landfills and prioritising recyclability. *"In the coming years, 10,000 to 15,000 wind turbines reaching the end of their life will pose a disposal problem. We expect to be able to guarantee recyclability in the future."*

Pressure vessels, automotive and sporting goods

Pressure vessels are another promising business area for Aditya Birla Chemicals, alongside the automotive and sporting goods industries, where recyclability is expected to become a key requirement.

The company has been working on the Recyclamine technology for over 10 years. This involves thermoset epoxy matrix and fibre reinforcement in composites that can be recovered, reused and repurposed. *"We were the first to develop this technology, acquiring it from the US company Connora Technologies. Our patented recycling process enables the recovery and reuse of composites made with Recyclamine. It is efficient and cost-effective with low energy consumption and operates at low to medium temperatures."* The properties of the recovered fibres are comparable to those of virgin fibres. *"We set up production facilities in Thailand which have been operational since the beginning of this year, with a capacity of 10,000 tons of hardener and 40,000 tons of resin – quite a large plant,"* Rajesh adds. Aditya Birla Chemicals was

Pressure vessels are a promising business area © Aditya Birla Group

The sporting goods industry is one of the major business fields for Aditya Birla Advanced Materials © Aditya Birla Group

the first company to offer this technology on a large scale, although there are now a few others operating in this field.

Two major players in the Indian wind industry are Inox and Suzlon in India, both customers of Aditya Birla Chemicals. Suzlon recently announced the launch of India's lowest carbon footprint wind turbine solution. The Suzlon Group is a leading global renewable energy solutions provider, with around 21.2 GW of wind energy capacity installed across 17 countries. The company has in-house R&D centres in Germany, the Netherlands, Denmark and India. *"At Suzlon, delivering world-class renewable energy solutions is rooted in the strength of our partnerships. Aditya Birla Advanced Materials has been a trusted collaborator for over two decades, playing a critical role in our journey. Their high-performance epoxy solutions, deep technical expertise, and spirit of co-innovation have consistently enabled us to manufacture wind turbine blades that meet the highest benchmarks of quality, reliability, and performance"*, J.P. Chalasani, CEO, Suzlon Group summarises the established relationship.

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Germany being a key market

Aditya Birla is also involved in several research projects, for example with Fraunhofer in Germany and Hanyang in Korea, as well as with the Defence Research and Development Organisation in India for specific applications. Germany is a key market for Aditya Birla Chemicals. Collaboration with leading European wind turbine manufacturers is managed from there. Rajesh explains: *"We have a composites application and R&D centre in Germany for various industry applications. 70% of our EU business comes from Germany. We are confident that Germany will overcome the multiple challenges in the long term."* In the medium term, Rajesh intends to increase the composites share of the business to 50%. *"We still want to grow in this segment. Globally, we want to definitively be at the top in order to serve the wind industry. In my opinion, there will be many opportunities in the future for companies leading on the front end of technology with sustainable solutions. Compared to other industries for epoxy, composites require much longer approval times, which enhances customer loyalty and our profitability."* Rajesh is quite confident that although momentum has slowed down temporarily largely due to demand related challenges, Europe will recover and he remains committed to the region.

Recyclability, a core issue

Aditya Birla Group, a US$67 billion (€57 billion) global conglomerate, is in the League of Fortune 500, engaged in 6 continents across 41 countries with a fine heritage of 165 years. The group is covering a wide range of sectors – from metals to cement, fashion to financial services, real estate, textiles and of course composites. Aditya Birla Chemicals, Advanced Materials has again been exhibiting at JEC World in Paris. Rajesh Balakrishnan thinks there will be many opportunities in the future for companies leading on the front end of technology with sustainable solutions. Recyclability is definitively a core issue in all activities of the advanced materials business and will become even more important in the coming years. □

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